Dyadic Packet-Transport Gauges

Natural Exact-Gram Lifts and the Geometry–Tail Coherence Barrier

Prime Dynamics Theory Program

July 2026

Abstract

RH-132 constructed the polar partial isometry between two given supports, but did not derive those supports from the model. Here the gauge is made dynamical and interscale. The frozen Gaussian models already carry a canonical dyadic isometry J from a coarse source-coordinate space to the next fine one. At each selected phase, the memory recursion and adaptive packet update produce an intrinsic four-frame U_σ. We align JU_σ to $U_{\sigma/2}$ by the overlap polar factor O_σ and lift it to the exact-Gram gauge

$$S_\sigma = G_\sigma^{-1/2} O_\sigma^* G_{\sigma/2}^{1/2}, \qquad S_\sigma^* G_\sigma S_\sigma = G_{\sigma/2}.$$

This is the first cross-scale gauge in the route determined by the model's grid and packet dynamics rather than by tail minimization.

The construction has a sharp limitation. Its tail factor is always at least the exact-Gram minimax factor, and no bound in terms of principal angles alone is possible: even identical packet frames can have natural to optimal tail-factor ratio $1/\varepsilon$. A sufficient replacement is a relative-tail coherence estimate in the natural frame.

The floor-free five-scale audit contains 96 phase-matched pairs. Thirty are zero-tail vacuous pairs, 24 are rank-birth pairs with infinite factor, and 42 have nonzero source and target tails. The natural gauge preserves 35 of the 37 positive transport-eligible pairs, hence 65 positive transfers overall versus 67 for the post-hoc optimum. On the 42 eligible pairs, however, its factor loss has median $10^{1.867} \approx 73.7$ and maximum $10^{9.987}$. The minimum principal cosine and logarithmic loss have sample correlation only -0.174. Thus the natural gauge is surprisingly viable for positivity, but a new tail-coherence law—not merely packet-angle coherence—is required for uniform theory.

1 From an abstract support map to a model-defined gauge

The source-coordinate widths in the frozen family double when σ is halved. The model construction uses the isometric coarse embedding

$$J_m e_j = 2^{-1/2}(e_{2j} + e_{2j+1}), \qquad J_m^* J_m = I.$$

This is not an invented comparison map: it is the same dyadic embedding used to form the coarse and detail blocks of the Gaussian operator. It therefore supplies a canonical identification between adjacent source-coordinate spaces.

At a fixed scale and time, let V be the recursively refreshed packet and let F be the leading four right singular frame of the projected recent cross action. The ambient input frame

$$U = VF$$

is orthonormal and depends only on the source seed, memory recursion, packet update, and chosen four-direction rule. For adjacent scales define

$$C = U^*_{\sigma/2} J U_\sigma = L \Sigma R^*.$$

The source-to-target polar alignment is $O_{st} = LR^*$ and the reduced target-to-source alignment is $O = O^*_{st}$, the standard polar/Procrustes choice for two frames [Golub and Loan, 2013].

2 Exact-Gram metric lift

The geometric frames and the action metrics contain different information. The polar factor aligns packet directions, while the positive Gramians $G_\sigma = A^*_\sigma A_\sigma$ measure their action size. They combine by a canonical metric lift.

Theorem 1 (Natural exact-Gram lift)**.** *Let $G, G' \succ 0$ and let O be any orthogonal target-to-source alignment. Then*

$$S = G^{-1/2} O (G')^{1/2}$$

satisfies $S^* G S = G'$. It is the unique exact-Gram lift whose normalized orthogonal factor is O. Moreover

$$|\det S| = \sqrt{\det G' / \det G}.$$

Proof. Direct multiplication gives $S^*GS = (G')^{1/2} O^* O (G')^{1/2} = G'$. Conversely, normalizing any exact-Gram gauge by $G^{1/2} S (G')^{-1/2}$ gives an orthogonal matrix, so fixing it to O fixes S. Taking determinants proves the last identity. □

Thus the dyadic packet geometry determines one distinguished member of the exact-Gram gauge family. Unlike the RH-121 gauge, it does not inspect the tail eigenspaces before choosing its orthogonal factor. The positive-matrix normalization and congruence identities are standard [Bhatia, 1997].

3 Tail cost and a sharp angle-only obstruction

Let

$$A = G^{-1/2} D G^{-1/2}, \qquad B = (G')^{-1/2} D' (G')^{-1/2}$$

be the normalized source and target tails. Under the natural lift, the least tail factor is the least b with

$$B \preceq b O^* A O.$$

Call it b_{nat}. The exact-Gram minimax factor b_* allows all orthogonal factors and therefore satisfies $b_* \le b_{\mathrm{nat}}$ by the ordered Loewner minimax principle [Horn and Johnson, 1991].

Proposition 2 (No principal-angle-only tail bound)**.** *For every $M > 0$ there are identical source and target packet frames, equal Gramians, and positive tails for which all principal cosines equal one, $b_* = 1$, but $b_{\mathrm{nat}} > M$.*

Proof. Take $G = G' = I_2$, identical frames, and hence $O = I_2$. Let

$$D = \mathrm{diag}(1, \varepsilon), \qquad D' = \mathrm{diag}(\varepsilon, 1).$$

The natural factor is $1/\varepsilon$. The orthogonal coordinate swap transports D exactly to D', so the minimax factor is one. Choosing $\varepsilon < 1/M$ proves the claim. □

The obstruction is stronger than poor principal-angle conditioning: it persists at angle zero. Packet geometry and tail eigendirections are two separate coherence problems.

Proposition 3 (Relative-tail coherence suffices)**.** *Suppose $A \succeq aI$ with $a > 0$ and*

$$\|B - O^* A O\| \le \delta.$$

Then

$$b_{\mathrm{nat}} \le 1 + \delta/a.$$

More generally, if $B \preceq \rho O^* A O + Q$ with $Q \succeq 0$, then the multiplicative coefficient is ρ and Q is an explicit normalized forcing term.

Proof. The norm bound gives $B \preceq O^* A O + \delta I$. Since $O^* A O \succeq aI$, one has $\delta I \preceq (\delta/a) O^* A O$. The affine statement is already in the desired Loewner form. □

Proposition 3 identifies the next source-level target: control the normalized tail discrepancy in the dyadic polar frame. It is strictly stronger than controlling projectors, but weaker and more natural than requiring the geometric gauge to be the post-hoc minimax gauge. Perturbation of such separated support data is governed by the usual gap mechanisms [Kato, 1995].

4 Five-scale common-assembly audit

We rebuild the same 120 states used in RH-130 with no positive Gram or tail floor. For each of the 96 adjacent-scale pairs, the coarse input frame is prolonged by the model's dyadic isometry, aligned to the target frame by polar SVD, and lifted at 90 decimal digits to an exact-Gram gauge. The largest recorded Gram alignment error is 3.73×10^{-18}.

The packet supports are not uniformly close. The minimum principal cosine ranges down to 0.001604, the median is 0.29434, and the maximum principal angle is 1.56919 radians, close to orthogonality. Nevertheless angle size does not predict tail-factor loss well. On the 42 nonzero-tail pairs, the logarithmic natural-to-optimal ratio has minimum 0.6765, median 1.8674, and maximum 9.9866; its sample correlation with the minimum principal cosine is -0.174.

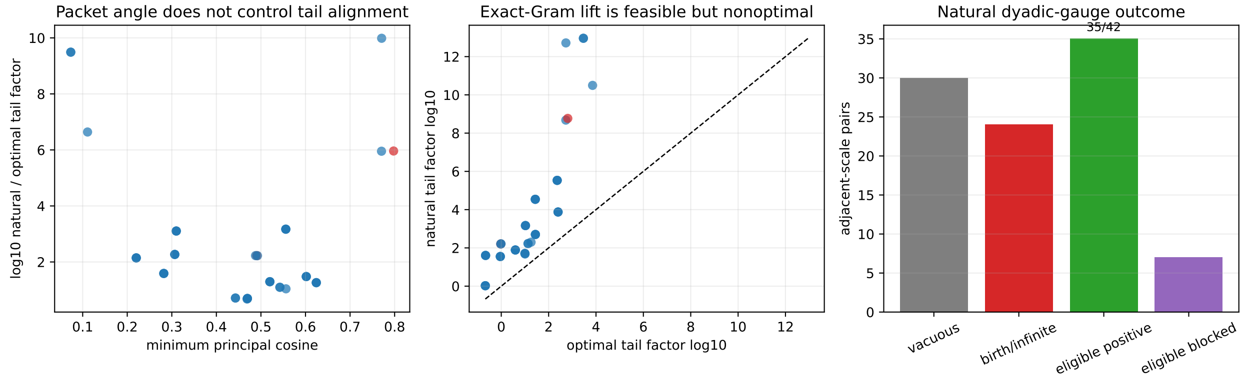

Figure 1: Natural versus optimal tail transport. Red points are the two pairs positive under the minimax gauge but blocked by the natural gauge.

The positivity count is much better than the factor ratios alone suggest. The 30 zero-to-zero pairs remain vacuously positive, and all 24 rank-birth pairs remain infinite, as they must. Of the 42 transport-eligible pairs, the post-hoc optimum is positive on 37 and the natural gauge on 35. Therefore the total count falls only from 67 to 65.

The two lost pairs are both left-channel final-phase cases. On $0.08 \to 0.04$ at threshold 10^{-4}, the minimum principal cosine is still 0.798, but the natural factor is 5.89×10^{8} versus optimal 643, producing $\gamma_{\mathrm{nat}} \approx 296$. On $0.04 \to 0.02$ at threshold 10^{-8}, the natural factor is 160.65 versus optimal 0.954, producing $\gamma_{\mathrm{nat}} \approx 3.95$. The first case is especially decisive: a visually coherent packet frame can be violently misaligned with the normalized tail.

5 Route consequence and boundary

RH-133 supplies a genuine model-derived cross-scale gauge. It uses the existing dyadic grid identification, the source-seeded packet recursion, and the four-direction frame; no tail eigenframe is consulted in its definition. It preserves nearly all finite positive transfers, so the natural-gauge route is not numerically dead. At the same time, the $10^{9.99}$ worst loss and Proposition 2 rule out any proof based only on principal angles.

The next paper should derive the tail discrepancy itself from the finite memory recursion. The desired identity is an affine decomposition of the target normalized tail into a dyadically transported old component plus a newly born memory slice and controlled frame-change terms. Those terms must feed explicit candidates for ρ_n and q_n.

We have not proved a uniform principal-angle law, a uniform natural tail factor, an all-level affine recurrence, a normalized-base liminf, uniform Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.

Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.